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PI Name:	Meiburg, Eckart H. Ph.D.		
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PI Email:	meiburg@engineering.ucsb.edu	**Fax:**	FY 805-893-5278
PI Organization Type:	UNIVERSITY	**Phone:**	805-893-5278
Organization Name:	University of California at Santa Barbara		
PI Address 1:	Department of Mechanical and Environmental Engineering		
PI Address 2:			
PI Web Page:	http://www.me.ucsb.edu/		
City:	Santa Barbara	**State:**	CA
Zip Code:	93106	**Congressional District:**	23
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Contact Monitor:	Balasubramaniam, R.	**Contact Phone:**	216-433-2878
Contact Email:	RBalasubramaniam@grc.nasa.gov		
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This project represents the computational component of a joint experimental and computational investigation into the dynamics of miscible interfaces. Within the computational component, it is proposed to carry out high resolution spectral element simulations that elucidate in significantly increased detail the role of 1) nontraditional Korteweg stresses and 2) non-vanishing divergence in miscible fluid flows. The geometry to be considered is the capillary tube also employed in the experiments.

The proposed research aims at obtaining fundamental insight into the importance of Korteweg stresses and effects of non-vanishing divergence. For this purpose we will continue the preliminary numerical simulations for different fluid pairs, with corresponding density differences and estimated Korteweg stress coefficients. Over the next year, these simulations will be analyzed in significantly more detail than we have been able to obtain so far. This will be

accomplished by employing high-accuracy spectral element methods with high resolution. By comparing the results of such simulations with those obtained for conventional systems (i.e., no Korteweg stresses and divergence free velocity field), we can obtain estimates for the importance of these effects in different miscible fluid combinations. In this way, the simulations will aid in the selection of optimally suited fluid pairs for the planned space experiments. Also, the discrepancies observed between the different simulations will guide experimental observations of flow features caused by the mechanisms under investigation.

Task Description:

This research has a practical motivation, since the improved understanding of the dynamics of multiphase porous media flows is a pre-requisite for progress in the fields of enhanced oil recovery, fixed bed regeneration, hydrology, filtration etc. From stability theory we know that if in a porous environment the displacing fluid is less viscous than the displaced fluid, the unfavorable mobility profile leads to the well known fingering instability. The displacing fluid subsequently channels through the displaced zone, leaving behind a layer of the more viscous fluid at the walls, thereby reducing the efficiency of the displacement process. When the fluids are immiscible, the interfacial tension (or the Capillary number, which is a measure of the ratio of viscous to surface tension forces) determines the most unstable wavelength of the fingering instability as well as the dynamics of the fingers, as they evolve.

When the liquids are miscible, a meniscus between the two is absent, and the interfacial tension tends to zero. Thus, at first sight, the dynamics appear to be simpler and are determined by the relative importance of convection and diffusion. However, the comparison of our ground based experiments and corresponding numerical simulations suggests that a number of secondary effects become important and influence the dynamics. The non-vanishing divergence of the velocity field and the Korteweg stresses, discussed earlier, are two such phenomena. In addition, the diffusion tensor at the interface may be anisotropic, and may depend on the local velocity field. While it is true that a large effective interfacial tension is unlikely to exist at a miscible interface, Joseph and coauthors have approached the stability problem based on the idea of such an effective interfacial tension. This approach was motivated by the observation that the form of miscible drops of one fluid moving in another is similar to that of immiscible fluids with a large interfacial tension. However, while it is easily possible to make such qualitative comparisons, it is much harder to establish them on sound quantitative footing. It is clear from the above discussion that miscible interfaces are rich in terms of dynamical phenomena that might influence its behavior. One output from this project would be an attempt to establish the proper form of the set of governing equations of motion for such flows. It is expected that these equations will be a modified form of the conventional Stokes equations, with additional or altered terms that allow for nonsolenoidal effects and Korteweg stresses. Under some circumstances, i.e. for some parameter regimes, we hope to be able to generate a suitable boundary condition to be used at a miscible interface, which will be compared with other approaches taken in the past.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Miscible displacements in Hele-Shaw cells: Two-dimensional base states and their linear stability:
Miscible fingering in a Hele-Shaw cell is studied by means of Stokes simulations and linear stability analysis. The two-dimensional simulations of miscible displacements in a gap indicate the existence of a quasisteady state near the tip of the displacement front for sufficiently large P\'{e}clet numbers and viscosity ratios, in agreement with earlier work by other authors. The front thickness of this quasisteady state is seen to scale with $Pe^{-1/2}$, while it depends only weakly on the viscosity ratio. The nature of the viscosity-concentration relationship is found to have a significant influence on the quasisteady state. For the exponential relationship employed throughout most of the investigation, we find that the tip velocity increases with Pe for small viscosity ratios, while it decreases with Pe for large ratios. In contrast, for a linear viscosity-concentration relationship the tip velocity is seen to increase with Pe for all viscosity ratios. The simulation results suggest that in the limit of high Pe and large viscosity contrast, the width and tip velocity of the displacement front asymptote to the same values as their immiscible counterparts in the limit of large capillary numbers.

In a subsequent step, the stability of the quasisteady front to spanwise perturbations is examined, based on the three-dimensional Stokes equations. For moderate values of Pe, the maximum growth rate is found to increase monotonically with the viscosity ratio. In contrast, for larger Pe the growth rate reaches a maximum for an intermediate viscosity ratio. The influence of Pe on the growth of the instability is also nonuniform. For mild viscosity contrasts, a larger Pe is found to be destabilizing, while for large viscosity contrasts an increase in Pe has a slightly stabilizing influence. A close inspection of the instability eigenfunction reveals the presence of two sets of counterrotating, roll-like structures, with axes aligned in the cross-gap and streamwise directions, respectively. The former lead to the periodic acceleration and deceleration of the front, while the latter result in the widening and narrowing of the front. These roll-like structures are aligned in such a way that the front widens where it speeds up, and narrows where it slows down. The findings from the present stability analysis are discussed and compared with their Darcy counterparts, as well as with experimental data by other authors for miscible and immiscible flows.

Variable density, miscible displacements in a vertical Hele-Shaw cell: linear stability:

A computational study based on the Stokes equations is conducted to investigate the effects of gravitational forces on miscible displacements in vertical Hele-Shaw cells. Nonlinear simulations provide the quasisteady displacement fronts in the gap of the cell, whose stability to spanwise perturbations is subsequently examined by means of a linear stability analysis. The 2D simulations indicate a marked thickening (thinning) and slowing down (speeding up) of the displacement front for flows stabilized (destabilized) by gravity. For the range investigated, the tip velocity is found to vary linearly with the gravity parameter. Strongly stable density stratifications lead to the emergence of flow patterns with spreading fronts, and to the emergence of a secondary needle-shaped finger, similar to earlier observations for capillary tube flows. In order to investigate the transition between viscously driven and purely gravitational instabilities, a comparison is presented between displacement flows and gravity driven flows without net displacements.

The linear stability analysis shows that both the growth rate and the dominant wave number depend only weakly on the Peclet number. The growth rate varies strongly with the gravity parameter, so that even a moderately stable density stratification can stabilize the displacement. Both the growth rate and the dominant wavelength increase with the viscosity ratio. For unstable density stratifications the dominant wavelength is nearly independent of the gravity parameter, while it increases strongly for stable density stratifications. Finally, the kinematic wave theory of \cite{laj-high} is seen to capture the stability limit quite accurately, while the Darcy analysis misses important aspects of

the instability.

Miscible displacements in a vertical Hele-Shaw cell: non-monotonic viscosity profiles:

The influence of nonmonotonic viscosity-concentration relationships on viscous fingering of neutrally buoyant, miscible fluids in a Hele-Shaw cell has been investigated. In a first step, quasisteady base states are obtained by means of nonlinear Stokes simulations. The properties of these base states are analyzed as a function of the Peclet number, the viscosity ratio, and the profile parameters. Subsequently, the stability of these base states is investigated by means of a linear stability analysis.

Overall, the nonmonotonicity of the viscosity-concentration relationship is seen to have a much smaller influence on Hele-Shaw displacements than on corresponding Darcy flows. The reason for this difference lies in the nature of the respective base states. For Darcy flows, the base state is characterized by constant velocity and a diffusively decaying concentration (and hence viscosity) profile. This base viscosity profile is strongly affected by the nonmonotonicity. On the other hand, for Hele-Shaw displacements the quasisteady base states are convectively dominated and characterized by sharp fronts, so that their shape depends only weakly on the details of the viscosity-concentration relationship. Hence, for Hele-Shaw displacements both the eigenfunctions and the associated growth rates are quite similar for monotonic and nonmonotonic profiles, in contrast to earlier findings for Darcy flows.

Variable density and viscosity, miscible displacements in horizontal Hele-Shaw cells: Linear stability analysis:

A computational investigation of variable density and viscosity, miscible displacements in horizontal Hele-Shaw cells is presented. As a first step, two-dimensional base states are obtained by means of nonlinear Stokes simulations. Here, the vertical position of the displacement front is seen to reach a quasisteady equilibrium value, reflecting a balance between viscous and gravitational forces. These base states allow for two instability modes: First, there is the familiar tip instability driven by the unfavorable viscosity contrast of the displacement, which is modulated by the presence of density variations in the gravitational field. Secondly, a gravitational instability occurs at the unstably stratified horizontal interface along the side of the finger. Both of these instability modes are investigated by means of a linear stability analysis. The gravitational mode along the side of the finger is characterized by a wavelength of about one half to one full gap width. It becomes more unstable as the gravity parameter increases, even though the interface is shifted closer to the wall. The growth rate is largest far behind the finger tip, where the interface is both thicker, and located closer to the wall, than near the finger tip. The competing influences of interface thickness and wall proximity are clarified by means of a parametric stability analysis.

The tip instability mode represents a gravity-modulated version of the neutrally buoyant mode. The analysis shows that in the presence of density stratification its growth rate increases, while the dominant wavelength decreases. This overall destabilizing effect of gravity is due to the additional terms appearing in the stability equations, which outweigh the stabilizing effects of gravity onto the base state.

Bibliography Type:	Description: (Last Updated: 04/27/2010)
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